Morphology as a Lever for Safe Reinforcement Learning: Constrained Co-Design of Body and Controller

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ABSTRACT

Safe reinforcement learning (RL) optimises a controller for a fixed body, yet whether a fast and safe behaviour exists, and whether it can be learned, depends on the body. We study constrained co-design, in which body and controller are optimised together under a safety budget. A Gaussian-mixture distribution over the limb and torso dimensions of a quadruped is evolved with score-function updates, while one morphology-conditioned PPO-Lagrangian controller, with a Lagrange multiplier per mixture component, is trained on a population of bodies simulated side by side on one GPU. On a hazard-field crossing task the co-designed robot is [55%] faster than the nominal one at a comparable safety cost. Ranking designs by performance alone selects unsafe bodies when the budget is hard to meet, so the design search needs constraint handling as well. Our MuJoCo Warp stack completes a 2-billion-step co-design run in under 3 hours on one consumer GPU.

CCS CONCEPTS

• **Computing methodologies** → **Reinforcement learning**; • **Computer systems organization** → **Evolutionary robotics**.

KEYWORDS

morphology–control co-design, safe reinforcement learning, constraint handling, evolutionary robotics, legged locomotion

1 INTRODUCTION

Legged robots increasingly have to move quickly through places where some contacts are unacceptable: crops they must not trample, clutter they must not disturb, hazards they must not step on. Safe RL formalises this as a constrained Markov decision process (CMDP): maximise return subject to a bound on expected cost [1, 2, 14]. It always optimises the controller of a *fixed* body. Yet the trade-off between safety and performance that a robot can reach belongs to body and controller together [13]: legs long enough to clear a hazard turn "weave around it" into "step over it".

Co-design work optimises the body for reward alone [6, 7, 16, 18], and safe-RL work keeps the body fixed [8, 10]. We found no work that trains the controller with constrained RL inside a design loop. We ask whether optimising morphology improves the safety–performance trade-off beyond what constrained policy optimisation reaches on the nominal body.

Adding a cost term to the co-design reward is not enough, because the constraint has to hold at two levels. Inside, the budget must hold for every body, not on average over the population the shared controller trains on. Outside, ranking designs by performance assumes the controller already meets the budget; when it does not, the search selects fast, unsafe bodies. The second is the classic constraint-handling problem of evolutionary computation [4, 5], with feasibility estimated from a controller that is still learning.

Teaser [render]: nominal (top) and co-designed (bottom) ant crossing the same hazard field at equal times.

Figure 1: Same task, same safety budget, different body.

Contributions.

(1) A safe co-design method: an evolved design distribution [16] with a morphology-conditioned PPO-Lagrangian controller, one Lagrange multiplier per mixture component, and [feasibility-first] design ranking.
(2) Evidence that morphology is a safety lever: the co-designed quadruped crosses a hazard field [55%] faster at comparable cost, and the evolved body changes with the hazards.
(3) A failure mode of naive safe co-design: speed-only ranking returns a body [5×] over budget when the constraint is hard to meet.
(4) An open GPU stack for safe RL and safe co-design on MuJoCo Warp, with a different body in every environment lane.

2 RELATED WORK

Safe RL.. Constrained RL is usually solved with Lagrangian or primal-dual methods such as PPO-Lagrangian [14, 20], or with trust-region and switching schemes [1, 19, 26]. Benchmarks include Safety Gym, Safety-Gymnasium and, on accelerators, mjx-safety-gym and CRAX [8, 14, 22, 23]. Constrained legged locomotion [3, 10] also fixes the body.

Co-design. Bodies have been evolved since virtual creatures [12, 18], and later jointly with RL controllers, through learned design distributions [7, 16], graph policies [24, 27] or Darwinian evolution with learning [6]. Where safety appears, it is a penalty in the reward [29], the task itself [28], or a design specification [9].

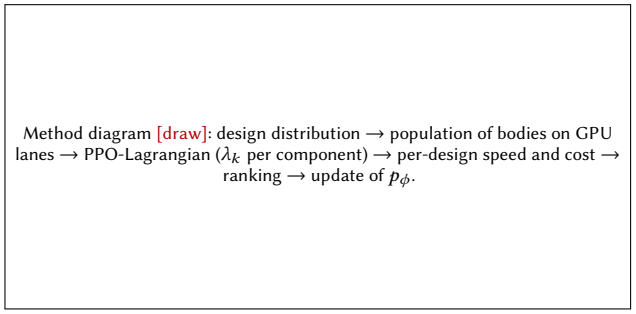


Figure 2: Constrained co-design.

Constraint handling in evolutionary computation. Penalties, feasibility rules [5], stochastic ranking [15] and ε-constrained methods [21]. Our design search belongs to the estimation-of-distribution / natural-evolution-strategy family [11, 25]; what is new is that feasibility is a noisy estimate that changes as the controller learns.

3 PROBLEM FORMULATION

A design $\xi \in \Xi = [0.6, 1.4]^7$ scales the lengths and radii of the three leg segments (shared by all four legs) and the torso radius of the Safety Gym ant [14]; the kinematic structure is fixed. For a design ξ, the controller solves a CMDP with episode cost J_C and budget d. Safe co-design is the bilevel problem

$$\begin{aligned} &\max_{\xi\in\Xi}\ v(\pi^*_\xi;\xi) \quad \text{s.t.} \quad J_C(\pi^*_\xi;\xi) \le d, \\ &\pi^*_\xi \in \arg\max_{\pi:\, J_C(\pi;\xi)\le d} J_R(\pi;\xi), \end{aligned} \tag{1}$$

where v is the crossing speed. As in Schaff et al. [16], one policy $\pi_\theta(a \mid s, \xi)$, conditioned on the design and trained on bodies drawn from a search distribution $p_\phi(\xi)$, solves the inner problem for the whole population at once. [One sentence on why speed, not return: return saturates once bodies arrive.]

4 METHOD

Population training on one GPU.. Each design iteration samples a population of designs from p_ϕ and gives each its own block of parallel environments. Per-lane physics models differ only in their geometry, inertia and actuator fields, so one compiled step function serves every body without recompilation. The design is appended to the observation.

Inner loop: per-component PPO-Lagrangian. PPO [17] with a separate cost critic is trained on the Lagrangian

$$\min_{\lambda\ge 0}\max_{\theta}\ \mathbb{E}_{k,\ \xi\sim p_\phi^k}\Big[J_R(\pi_\theta;\xi) - \lambda_k\big(J_C(\pi_\theta;\xi) - d\big)\Big], \tag{2}$$

with one multiplier λ_k per mixture component, so that the budget holds for each subpopulation rather than on average. Multipliers are capped to avoid integral windup.

Outer loop: evolving the design distribution. Following Schaff et al. [16], p_ϕ is a mixture of 8 diagonal Gaussians with frozen uniform weights, updated by a score-function gradient through each sample's own component,

$$\nabla_\phi \approx \tfrac{1}{M}\textstyle\sum_{i=1}^{M} u_i\, \nabla_\phi \log p_\phi^{k_i}(\xi_i), \tag{3}$$

where u_i are the designs' standardised scores; as in the reference implementation, each design's gradient is clipped to norm 0.5 before averaging. The distribution commits to one design by repeatedly retiring the worst half of its components, each scored on 100 freshly drawn designs under the current policy, and the controller is fine-tuned on the final mode (Algorithm 1).

Algorithm 1 Constrained co-design

```
F ← 8 random families of body designs
π ← new controller;  λ_f ← small safety penalty for each family f
for each iteration do
    B ← bodies sampled from F
    for all bodies b ∈ B in parallel do
        run π on b; record its speed, cost and arrivals
    end for
    update π: maximise speed − λ_f · cost
    rank B: safe, then unsafe, then not arriving
    for all families f ∈ F do
        if cost of f's bodies > budget then
            raise λ_f
        else
            lower λ_f
        end if
        move f towards its best-ranked bodies
    end for
    if prune step then
        remove the worst half of F
    end if
end for
return the best body of the last family, π
```

Constraint-aware ranking. Designs are ranked feasibility first [5]. A design is feasible if its mean episode cost is within the budget and it reaches the goal in at least half of its episodes. Feasible designs rank by speed, designs that arrive but exceed the budget rank by their cost excess, and designs that do not arrive rank last. Arrival is a constraint of higher priority than cost, so a body that never moves, and therefore never pays cost, cannot win. The update uses rank utilities [25]. Because a chop is irreversible, a component counts as over budget there only when its cost exceeds the budget by more than two standard errors.

5 EXPERIMENTAL SETUP

Task. The ant crosses an 11 m corridor containing 20 hazard discs (radius 0.16 m) on a staggered lattice, or redrawn at random at every reset. Hazards do not collide; touching one with a grounded body part costs up to 1 per step, graded by penetration as in Safety Gym. A flip or a corridor exit ends the episode with a cost of 50. The budget is $d = 2$ per episode. A zero-cost crossing exists for the nominal ant [cite kinematic check].

Conditions. All on the random minefield, the setting where the budget binds, with 3 seeds each: the nominal ant with PPO-Lagrangian; co-design with speed ranking; co-design with feasibility-first ranking; each evolved body retrained from scratch

Table 1: Main hyperparameters.

Environments	4096 (64 designs × 64 lanes)
Policy / value networks	MLP 256 × 4 / 256 × 5
PPO lr, minibatches × size	2×10^{-4}, 32 × 128
Multiplier lr / cap	6×10^{-5} / 3.0
Design components / lr	8 / 0.08
Schedule (co-design)	2B steps; warm-up 200M; chops 400/600/800M

[run]: speed against episode cost for the nominal ant and co-designed bodies across budgets.

Figure 3: Safety–performance frontier.

on its own; unconstrained co-design; and the unconstrained body retrained with the constraint. [Runs to complete.]

Evaluation. 512 stochastic episodes per checkpoint on held-out layouts. We report return, speed, episode cost, hazard steps, and the shares of flipped and out-of-bounds episodes. Hyperparameters are in Table 1.

6 RESULTS

[All numbers below are preliminary: one seed, earlier simulation stack, and co-design code before the 2026-10-19 fixes. Regenerate everything.]

Morphology improves the trade-off. At budget 2 the co-designed ant crosses at 1.89 m/s, against 1.20–1.22 m/s for the nominal ant at comparable cost (Table 2). [Frontier over budgets (Fig. 3); controls for training budget and seeds.]

What evolves. Long legs at the box limit and a small torso evolve on every hazard regime; shin length shrinks as hazards get larger or random (1.40, 1.22, 1.03). [Gene-trajectory figure; gait analysis.]

The design search needs constraint handling. On random fields the inner loop could not meet the budget for any component, and speed-only ranking kept the fastest, most violating body (cost 10.15). [Feasibility-first result.]

Simulation stack. On an RTX 3090, MuJoCo Warp trains 3.1–3.3× faster than the MJX-JAX implementation it extends, and a 2B-step co-design run takes 2.6 h at 4096 environments instead of 13.9 h. Physics agrees with MJX-JAX to about 10^{-7} per step. [Same-task comparison with CRAX.]

7 DISCUSSION AND LIMITATIONS

The nominal ant can cross at zero cost, but constrained RL does not find that precise foot placement; a body with more clearance and a longer stride makes the safe crossing easier to learn. Morphology may therefore help safety mainly through learnability rather than feasibility [test]. Limitations: few seeds; several genes end at the box limit; actuators are rescaled with body size, so size is free; on this task speed and safety pull the same way; and the design space is parametric, because GPU simulators batch one kinematic structure.

8 CONCLUSION

Safety is a property of the body as well as of the controller. Co-designing morphology under a safety constraint produced a faster quadruped at comparable cost, and it needs constraint handling at both levels: per-component multipliers in the controller and constraint-aware ranking in the design search.

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Table 2: Evaluation at budget $d = 2$ (lattice field unless noted). [preliminary; regenerate.]

Body	Training	Return	Speed (m/s)	Cost	Hazard steps	Flipped (%)	OOB (%)
Nominal	PPO-Lag, $d = 0$	11.01	1.20	1.9	3.5	1.4	1.1
Nominal	PPO-Lag, $d = 5$	10.99	1.22	2.7	5.1	2.1	1.1
Nominal	PPO-Lag, $d = 2$	[]					
Co-designed	co-design	11.09	1.89	2.55	6.6	0.4	1.8
Co-designed, random field	co-design, speed ranking	10.88	1.48	10.15	25.8	1.4	4.9
Co-designed, random field	co-design, feasibility first	[]					

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